

SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Computer Network

COURSE CODE: CSA013

COURSE OUTCOME COVERED

CO1: Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)

Assessment Tool Weightage: 30%

Annexure A
CONCEPT MAPPING

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Code of Conduct:

I Suganthan Kumaresan (192424105) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

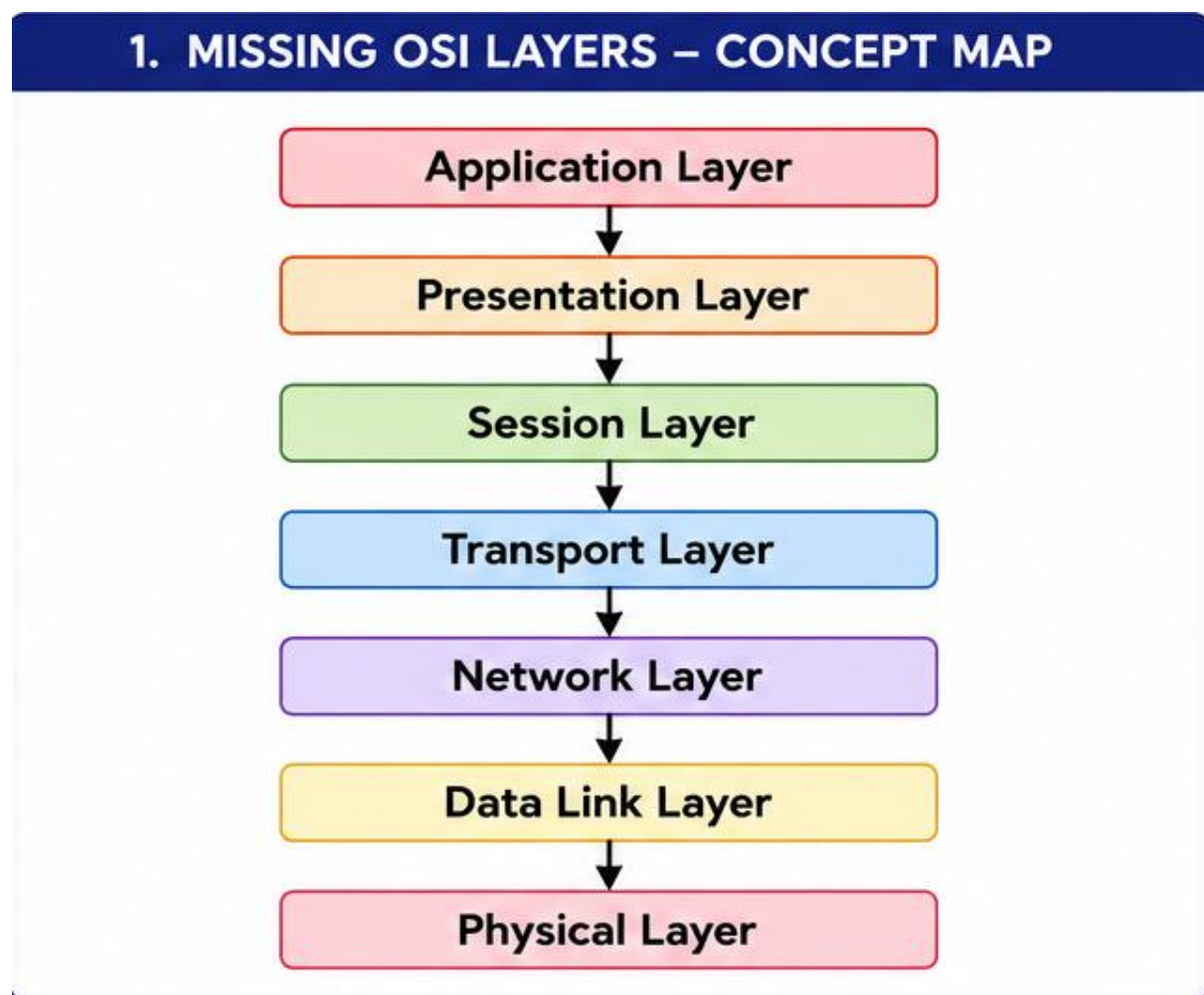
K. Suganthan

1. A partially completed concept map is given below: Application → Presentation → _____ → Transport → Network → _____ → Physical. Analyze the missing layers and explain how their functions are essential for reliable communication.

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Answer

Completed Concept Map



Missing Layers

- Session Layer
- Data Link Layer

Explanation

Session Layer

- Establishes, manages, and terminates communication sessions between applications.
- Provides synchronization through checkpoints.
- Allows communication to resume from the last checkpoint if a failure occurs.

Data Link Layer

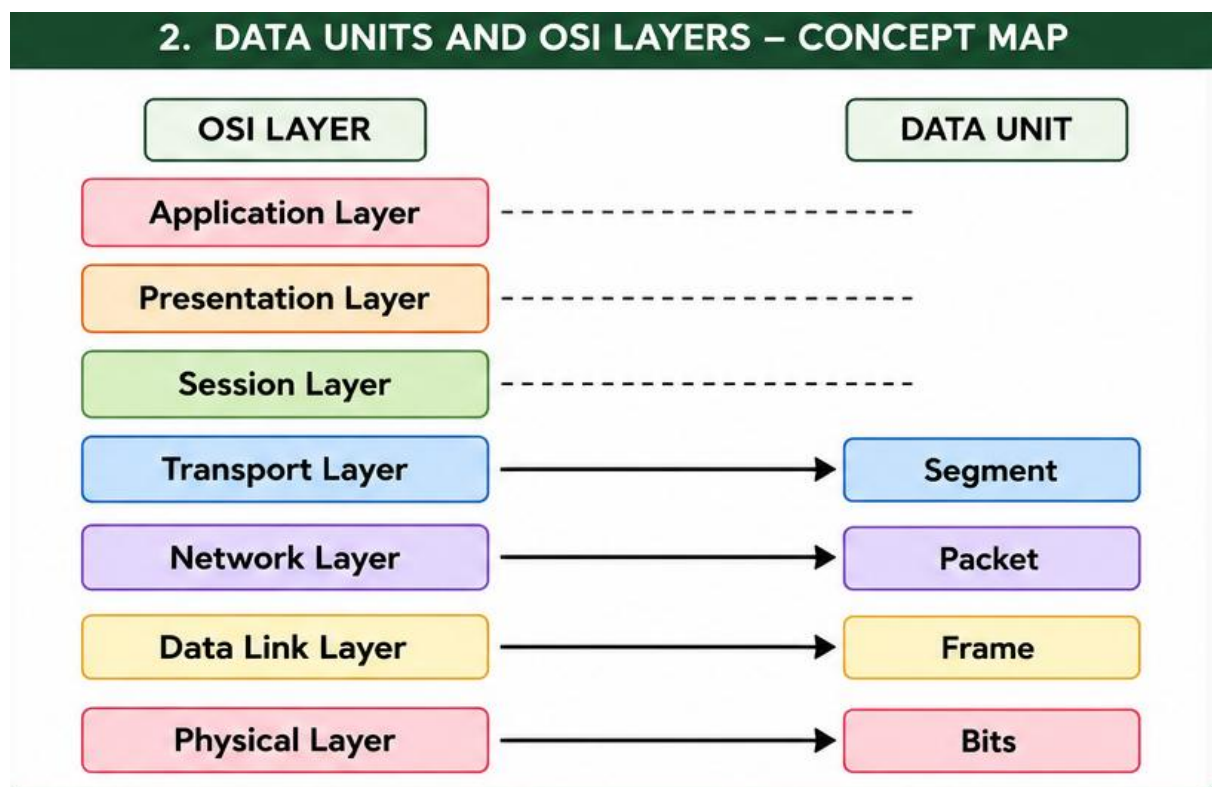
- Converts packets into frames.
- Uses MAC addresses for local delivery.
- Detects transmission errors using CRC (Cyclic Redundancy Check).
- Controls access to the transmission medium.

Conclusion

The **Session Layer** ensures smooth communication by managing sessions and synchronization, while the **Data Link Layer** provides reliable node-to-node communication through framing, error detection, and local addressing.

2. Construct a concept map linking data units (Segment, Packet, Frame, Bits) with their respective OSI layers and analyze how transformation of data units supports transmission.

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Analysis

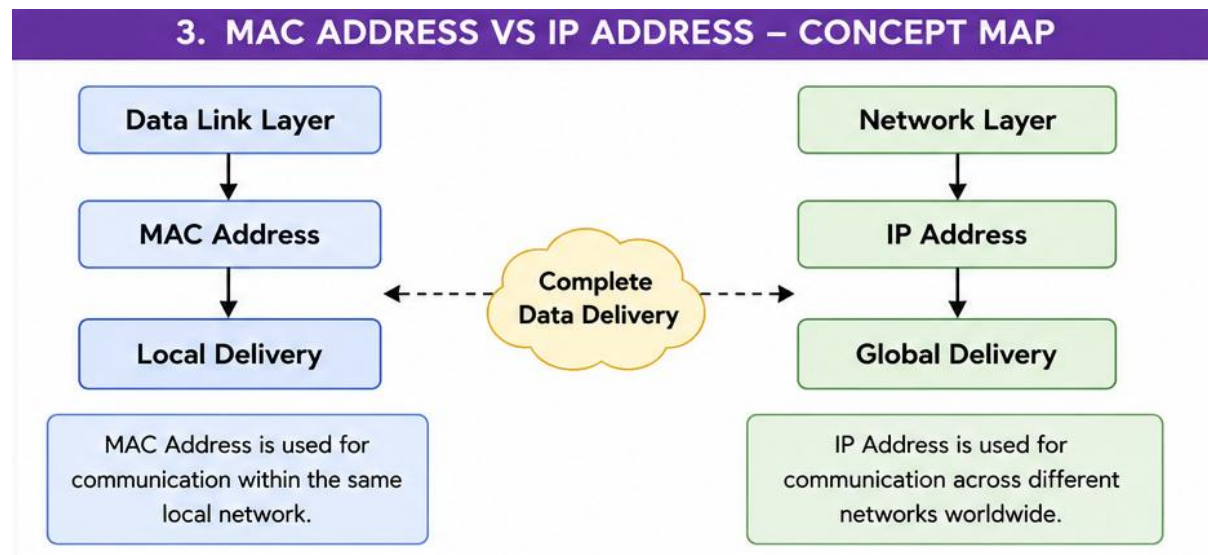
- **Transport Layer → Segment**
 - Divides application data into manageable segments.
 - Adds sequence numbers and port numbers.
- **Network Layer → Packet**
 - Encapsulates segments into packets.
 - Adds logical (IP) addresses for routing.
- **Data Link Layer → Frame**
 - Encapsulates packets into frames.
 - Adds MAC addresses and error detection information.
- **Physical Layer → Bits**
 - Converts frames into binary bits.
 - Transmits bits through the physical medium.

Conclusion

The transformation of data units allows each OSI layer to add the required information for reliable communication, routing, error detection, and physical transmission.

3. Given the concept map: DataLink→MACAddress→LocalDelivery Network → IP Address → Global Delivery Explain how these relationships ensure complete data delivery across networks.

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Explanation

Data Link Layer

- Uses **MAC addresses** to identify devices within the same local network.
- Ensures data reaches the correct device on the local network.

Network Layer

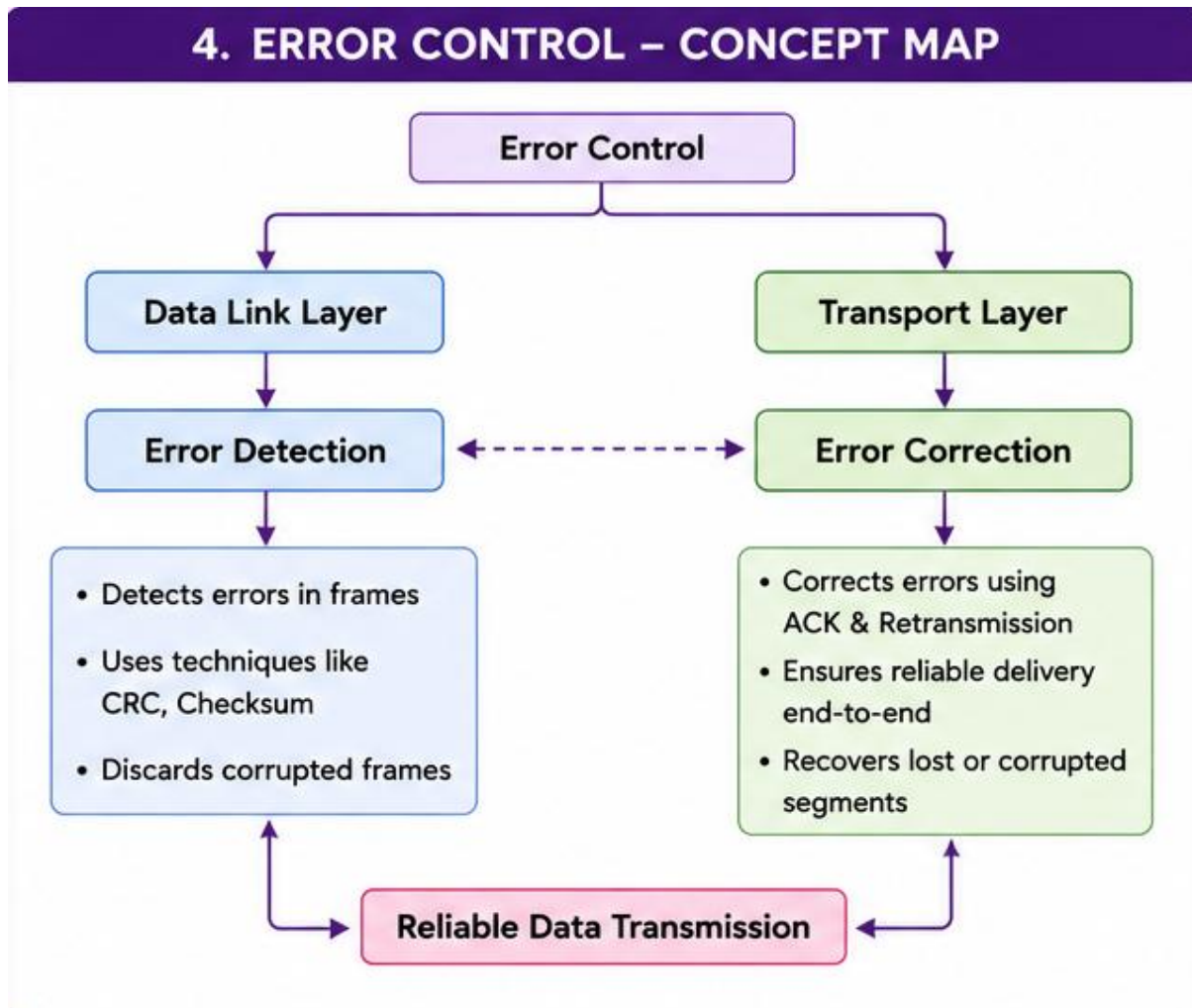
- Uses **IP addresses** to identify devices across different networks.
- Determines the best route from source to destination.

Conclusion

The **Data Link Layer** provides local delivery using MAC addresses, while the **Network Layer** provides global delivery using IP addresses. Together, they ensure that data reaches the correct destination across interconnected networks.

4. A concept map shows: ErrorControl → DataLinkLayer → ErrorDetection → Transport Layer → Error Correction Analyze how the interaction between these layers improves reliability in data transmission.

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Analysis

Data Link Layer

- Detects transmission errors using CRC and checksums.
- Identifies corrupted frames before forwarding them.

Transport Layer

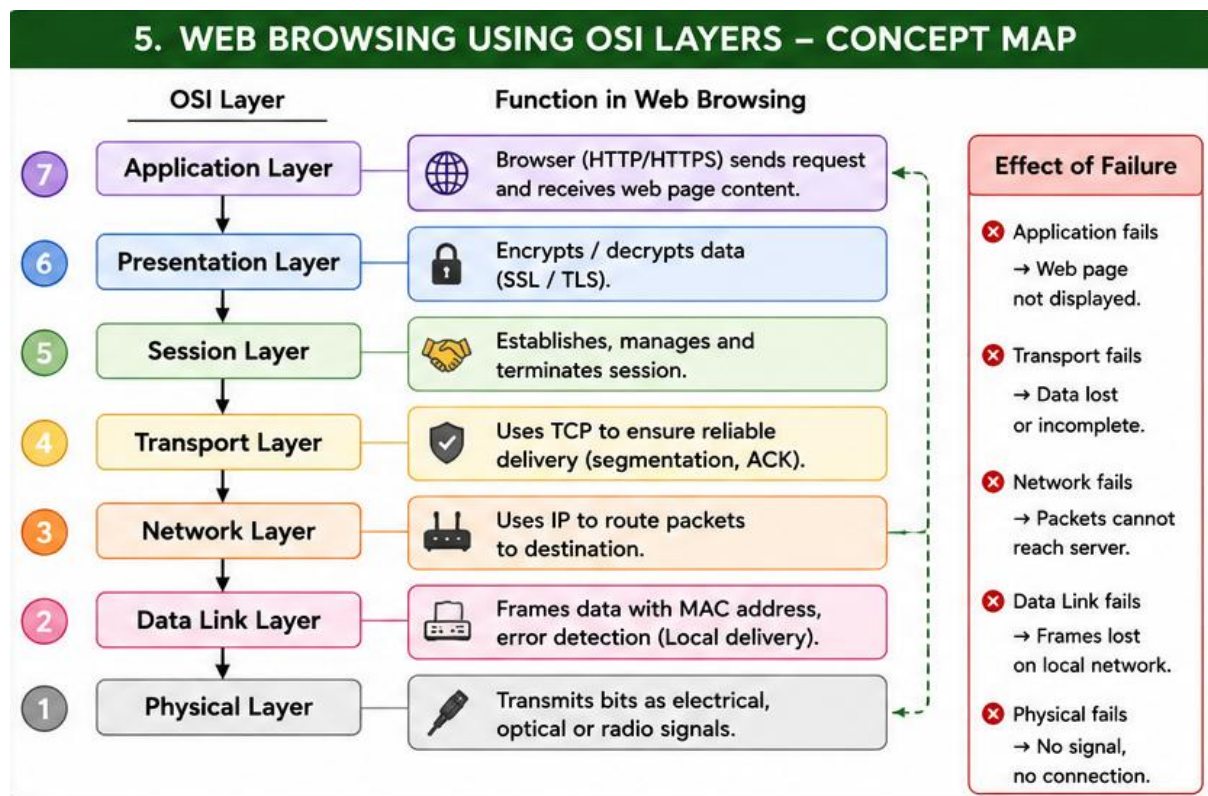
- Detects missing or lost segments.
- Uses acknowledgements (ACKs), sequence numbers, and retransmissions to correct errors.

Conclusion

The Data Link Layer detects transmission errors, while the Transport Layer corrects errors through retransmission. Together, they provide reliable end-to-end communication.

5. Develop a concept map for web browsing using OSI layers and analyze how failure in one layer affects the entire communication process.

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Analysis

- **Application Layer:** Requests web pages using HTTP/HTTPS.
- **Presentation Layer:** Encrypts data using SSL/TLS and compresses information.
- **Session Layer:** Establishes and maintains communication sessions.
- **Transport Layer:** Ensures reliable delivery using TCP.
- **Network Layer:** Routes packets using IP addresses.
- **Data Link Layer:** Delivers frames within the local network.
- **Physical Layer:** Transmits bits over cables or wireless signals.

Effect of Layer Failure

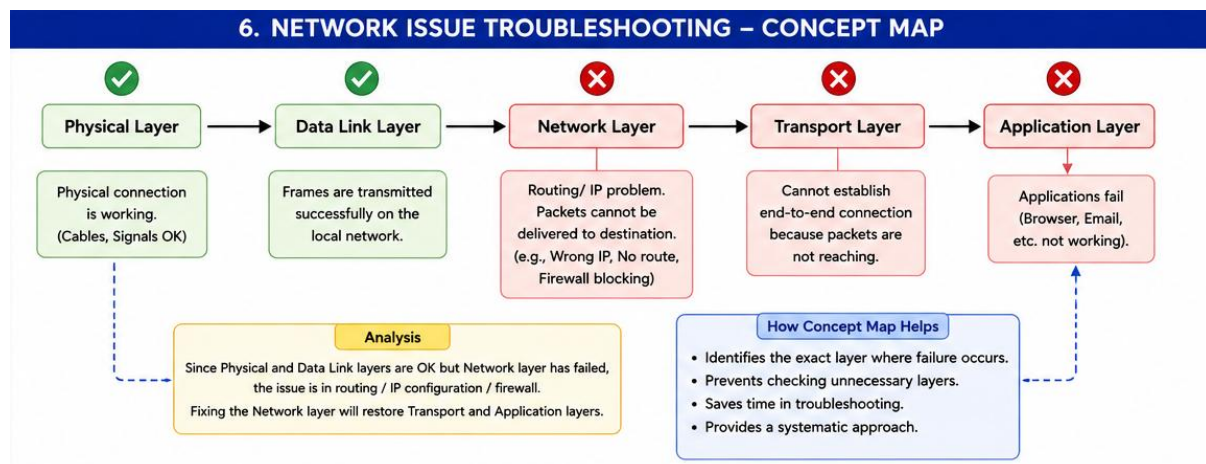
- Failure of the **Physical Layer** stops all communication.
- Failure of the **Data Link Layer** prevents local frame delivery.
- Failure of the **Network Layer** prevents routing between networks.
- Failure of the **Transport Layer** causes unreliable communication.
- Failure of the **Application Layer** prevents users from accessing websites.

Conclusion

Since each OSI layer depends on the services of the layer below it, failure in any layer disrupts the entire communication process.

6. A network issue is represented in the concept map: Physical ✓ → Data Link ✓ → Network ✗ → Transport ✗ → Application ✗ Analyze the problem and explain how the concept map helps in troubleshooting.

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Analysis

- **Physical Layer** ✓
 - The network cable or wireless connection is functioning correctly.
- **Data Link Layer** ✓
 - Frames are successfully transmitted within the local network.
- **Network Layer** ✗
 - Routing or IP addressing has failed.

- Possible causes include incorrect IP configuration, routing failures, or network congestion.
- **Transport Layer ✕**
 - Since packets cannot reach the destination, TCP/UDP communication fails.
- **Application Layer ✕**
 - Network applications such as web browsers, email, and FTP cannot communicate because the lower layers have failed.

How the Concept Map Helps in Troubleshooting

The concept map clearly shows that the **Physical** and **Data Link** layers are working, while the failure begins at the **Network Layer**. This helps network administrators quickly identify the source of the problem, reducing troubleshooting time and ensuring faster resolution.

Conclusion

The network issue originates at the **Network Layer**, causing failures in all higher layers. Concept maps provide a visual representation of dependencies between OSI layers, making fault isolation and troubleshooting more efficient.